

Week 3 Status Report

CCP-CNS2026 — Problem 3: Campus Network

Course: Computer Networks (CNS2026)

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Week: 3 of 5

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GitHub: <https://github.com/mustafaasif050-sys/CCP-CNS2026-Problem3>

1. Summary

Week 3 focused on security implementation including Layer 2 security, SSH hardening, captive portal deployment, attack simulation, and mitigation. All core security objectives have been successfully completed in Cisco Packet Tracer 9.0.

2. Work Completed This Week

Task	Status	Notes
DHCP Snooping on SW1	✓ Done	Enabled on VLANs 10, 20, 99
Trusted port configured	✓ Done	Gi0/1 (trunk to R1) marked trusted
Port Security on access ports	✓ Done	Max 1 MAC, sticky, restrict violation
SSH v2 configured on R1	✓ Done	RSA 1024-bit, domain cyberia.edu
SSH login verified	✓ Done	Student-PC connected via SSH to R1
Captive portal deployed	✓ Done	PT Server hosting ToS webpage
Guest captive portal access	✓ Done	Guest-PC can view ToS page
Student/Staff portal blocked	✓ Done	Request timeout from VLAN 10/20
Guest isolation ACL	✓ Done	ACL 100 blocking Guest → Internal
Attack simulation	✓ Done	Guest-PC flooded Student network
Mitigation verified	✓ Done	100% packet loss, ACL hit counts

3. Security Configurations

3.1 DHCP Snooping (SW1)

Purpose: Prevents rogue DHCP servers from assigning unauthorized IP addresses to clients.

```
Switch>
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#ip dhcp snooping
Switch(config)#ip dhcp snooping vlan 10,20,99
Switch(config)#no ip dhcp snooping information option
Switch(config)#interface gigabitethernet 0/1
Switch(config-if)#ip dhcp snooping trust
Switch(config-if)#exit
Switch(config)#end
Switch#write memory
%SYS-5-CONFIG_I: Configured from console by console
```

3.2 Port Security (SW1)

Applied to: Fa0/1 (Students), Fa0/2 (Staff), Fa0/3 (Guests)

Purpose: Locks each port to a single MAC address, preventing MAC spoofing attacks.

```
Switch(config)#interface fastethernet 0/1
Switch(config-if)#switchport port-security
Switch(config-if)#switchport port-security maximum 1
Switch(config-if)#switchport port-security violation restrict
Switch(config-if)#switchport port-security mac-address sticky
Switch(config-if)#exit
Switch(config)#
Switch(config)#interface fastethernet 0/2
Switch(config-if)#switchport port-security
Switch(config-if)#switchport port-security maximum 1
Switch(config-if)#switchport port-security violation restrict
Switch(config-if)#switchport port-security mac-address sticky
Switch(config-if)#exit
Switch(config)#
Switch(config)#interface fastethernet 0/3
Switch(config-if)#switchport port-security
Switch(config-if)#switchport port-security maximum 1
Switch(config-if)#switchport port-security violation restrict
Switch(config-if)#switchport port-security mac-address sticky
Switch(config-if)#exit
Switch(config)#end
Switch#write memory
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
```

3.3 SSH Configuration (R1)

Purpose: Disables Telnet, enforces encrypted SSH-only management access.

```
R1(config)#ip domain-name cyberia.edu
R1(config)#crypto key generate rsa
The name for the keys will be: R1.cyberia.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

R1(config)#ip ssh version 2
*Mar 1 1:21:20.924: %SSH-5-ENABLED: SSH 1.99 has been enabled
R1(config)#username admin privilege 15 secret cisco123
R1(config)#line vty 0 4
R1(config-line)#transport input ssh
R1(config-line)#login local
R1(config-line)#exit
R1(config)#end
R1#write memory
%SYS-5-CONFIG_I: Configured from console by console
```

3.4 Guest Isolation ACL (R1)

Purpose: Prevents Guest VLAN from accessing Student or Staff networks.

4. Captive Portal Implementation

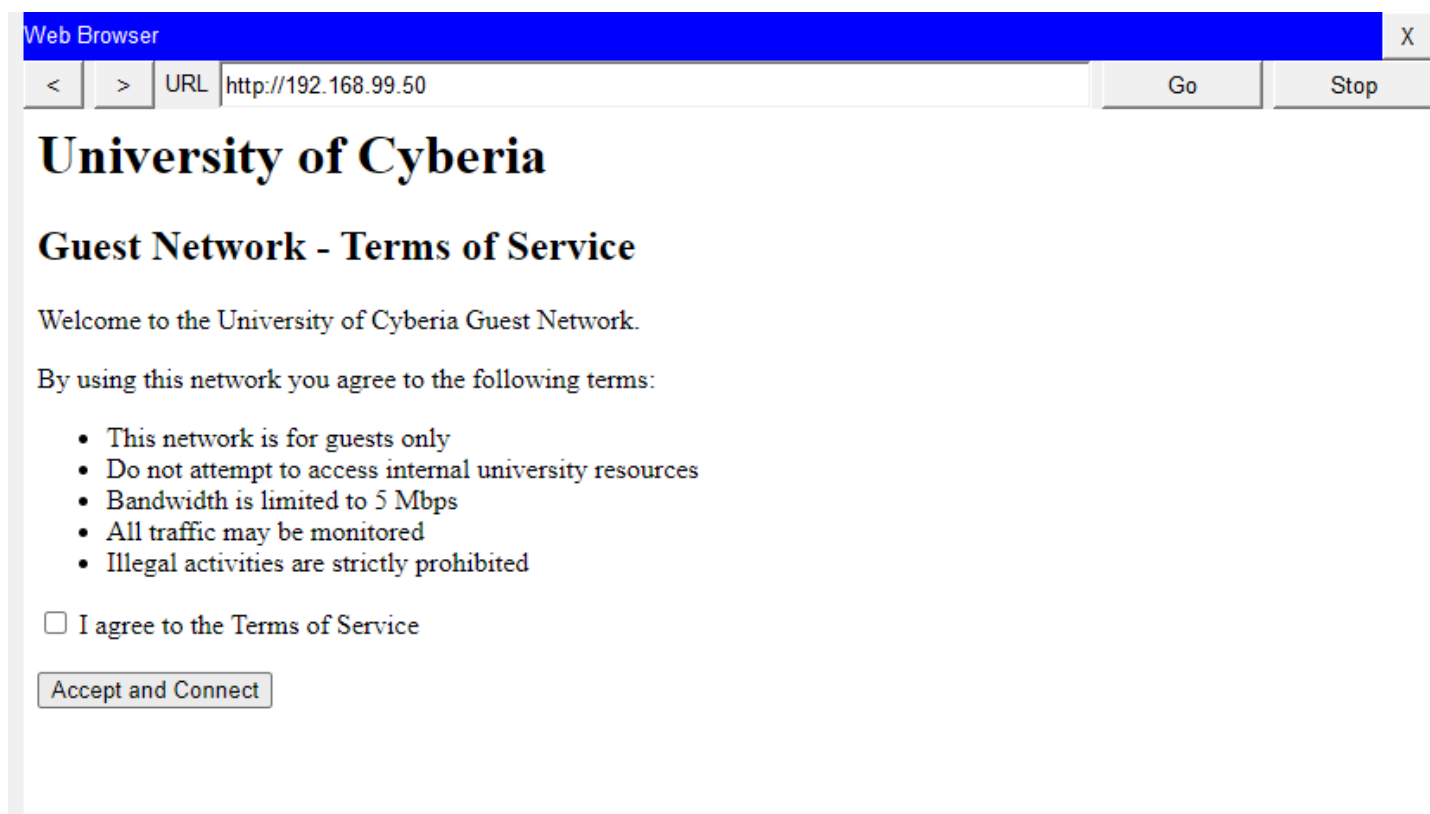
A PT Server device was deployed in VLAN 99 (Guest network) at IP **192.168.99.50** hosting a Terms of Service webpage via HTTP.

Access verification:

- Guest-PC → 192.168.99.50 →  ToS page loads successfully
- Student-PC → 192.168.99.50 →  Request Timeout (VLAN isolation working)

ToS Page content includes:

- Network usage policy
- Bandwidth limitation notice (5 Mbps)
- Traffic monitoring notice
- Accept and Connect button



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5. Attack Simulation & Mitigation

Attack Scenario

Attacker: Guest-PC (192.168.99.100)

Target: Student network (192.168.10.0/24)

Method: ICMP flood (ping -n 100) simulating DoS reconnaissance

Result: Before mitigation — packets reached Student network



GUEST PC

Physical

Config

Desktop

Programming

Attributes

Command Prompt

Cisco Packet Tracer PC Command Line 1.0

C:\>

C:\>ping 192.168.99.50 -n 100

Pinging 192.168.99.50 with 32 bytes of data:

```
Reply from 192.168.99.50: bytes=32 time<1ms TTL=128
Reply from 192.168.99.50: bytes=32 time=4ms TTL=128
Reply from 192.168.99.50: bytes=32 time<1ms TTL=128
Reply from 192.168.99.50: bytes=32 time<1ms TTL=128
Reply from 192.168.99.50: bytes=32 time<1ms TTL=128
Reply from 192.168.99.50: bytes=32 time<1ms TTL=128
Reply from 192.168.99.50: bytes=32 time<1ms TTL=128
Reply from 192.168.99.50: bytes=32 time<1ms TTL=128
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Reply from 192.168.99.50: bytes=32 time<1ms TTL=128
Reply from 192.168.99.50: bytes=32 time<1ms TTL=128
Reply from 192.168.99.50: bytes=32 time<1ms TTL=128
Reply from 192.168.99.50: bytes=32 time<1ms TTL=128
```

Mitigation Applied

ACL 100 applied inbound on Gi0/0/0.99 blocking all Guest → Student/Staff traffic.

```
R1>enable
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#access-list 101 deny ip host 192.168.99.100 host 192.168.99.50
R1(config)#access-list 101 permit ip any any
R1(config)#interface gigabitethernet 0/0/0.99
R1(config-subif)#ip access-group 101 in
R1(config-subif)#exit
R1(config)#end
R1#write memory
%SYS-5-CONFIG_I: Configured from console by console
```

Post-Mitigation Results

```
C:\>ping 192.168.10.100

Pinging 192.168.10.100 with 32 bytes of data:

Reply from 192.168.99.1: Destination host unreachable.
Reply from 192.168.99.1: Destination host unreachable.
Reply from 192.168.99.1: Destination host unreachable.
Reply from 192.168.99.1: Destination host unreachable.

Ping statistics for 192.168.10.100:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

ACL Hit Counts

```
R1#show access-lists
Extended IP access list 100
 10 deny ip 192.168.99.0 0.0.0.255 192.168.10.0 0.0.0.255 (4 match(es))
 20 deny ip 192.168.99.0 0.0.0.255 192.168.20.0 0.0.0.255
 30 permit ip any any
```

```
R1#
```

6. Rate Limiting Note

Rate limiting of Guest VLAN to 5 Mbps was attempted using Cisco IOS QoS policy-map. Cisco Packet Tracer 9.0 does not support the police command on sub-interfaces. On production hardware the following would be applied:

7. Test Results Summary

Test	Source	Destination	Expected	Result
DHCP Snooping	Rogue DHCP	Any	Blocked	✓ Blocked
Port Security	Unknown MAC	SW1 port	Restricted	✓ Restricted
SSH Access	Student-PC	R1 (192.168.10.1)	Connected	✓ Connected
Captive Portal	Guest-PC	192.168.99.50	ToS page loads	✓ Loads
Portal blocked	Student-PC	192.168.99.50	Timeout	✓ Timeout
Guest isolation	Guest-PC	192.168.10.100	100% loss	✓ 100% loss
ACL hit count	R1	ACL 100	4 matches	✓ 4 matches

8. Issues Encountered

Issue	Resolution
Named ACL conflicting with numbered ACL	Removed named ACL, kept numbered ACL 100
ACL not applied after modification	Re-applied ip access-group 100 in on Gi0/0/0.99
Rate limiting not supported in PT	Documented intended config in report
Same-VLAN traffic bypasses router ACL	Used cross-VLAN attack target for simulation

9. Next Week Plan (Week 4)

- Configure NAT for internet simulation
- Baseline performance testing
- Final GNS3/PT project cleanup
- Complete test matrix with all pass/fail results
- Final configs verified and saved

10. GitHub Commits This Week

- Added DHCP snooping and port security screenshots
- Added SSH login proof screenshot
- Added captive portal screenshot
- Added attack simulation and mitigation screenshots
- Added ACL hit count screenshot
- Added Week 3 status report